

Contents

Problem Statement	4
Business Context	4
Objective.....	4
Data Description	4
Data Overview	5
Exploratory Data Analysis	5
Univariate Analysis.....	5
Average Credit Limit	6
Total Credit Cards	7
Total Visits to Bank	7
Total Visits Online.....	8
Total Calls Made	9
Bivariate Analysis	9
Correlation Heatmap.....	10
Pairplot of Numerical Features	11
Avg Credit Limit vs Total Credit Cards	12
Total Credit Cards vs Total Calls Made	12
Total Visits Bank vs Total Visits Online.....	13
Avg Credit Limit vs Total Calls Made	14
Data Preprocessing.....	14
Missing Value Treatment.....	14
Outlier Detection and Treatment.....	14
Feature Engineering	15
Data Scaling	15
K-Means Clustering	15
Elbow Curve.....	16
Silhouette Score Analysis	16
Silhouette Visualizer.....	17
K-Means Cluster Profiling.....	18
Hierarchical Clustering	20
Cophenetic Correlation Analysis.....	20
Dendrograms for All Linkage Methods	22
Ward Dendrogram — Optimal Number of Clusters	23

Silhouette Score Validation	24
Hierarchical Clustering — Cluster Profiling	24
K-Means vs. Hierarchical Clustering — Comparison	25
Actionable Insights and Recommendations	27

Problem Statement

Business Context

AllLife Bank wants to focus on its credit card customer base in the next financial year. The marketing research team has advised that market penetration can be improved through personalized campaigns targeting new customers as well as upselling to existing customers. Additionally, customer feedback indicates that support services are perceived as poor, and the Operations team wants to upgrade the service delivery model to resolve customer queries faster.

Objective

The objective of this engagement is to apply unsupervised machine learning techniques, specifically clustering algorithms, to segment AllLife Bank's existing credit card customer base into distinct, meaningful groups based on their financial attributes and interaction behaviour with the bank.

The segmentation is expected to achieve the following outcomes:

- Identify natural customer groups that share similar spending patterns, credit profiles, and channel preferences.
- Profile each segment in terms of its defining characteristics, size, and business significance.
- Provide actionable, segment-specific recommendations to the Marketing team on how to personalize campaigns and improve targeting effectiveness.
- Provide actionable, segment-specific recommendations to the Operations team on how to prioritize service model improvements and reduce customer friction for the most affected groups.

The end deliverable is a data-driven segmentation framework that enables both the Marketing and Operations teams to move from generic, one-size-fits-all strategies to precision-targeted interventions — improving customer satisfaction, increasing product penetration, and reducing service costs simultaneously.

Data Description

The dataset contains information about 660 credit card customers of AllLife Bank and their financial attributes and banking interaction behaviour.

Column Name	Description
SI_No	Primary key of the records
Customer Key	Customer identification number
Avg_Credit_Limit	Average credit limit of each customer for all credit cards
Total_Credit_Cards	Total number of credit cards possessed by the customer

Total_visits_bank	Total number of visits the customer made (yearly) to the bank in person
Total_visits_online	Total number of online logins made by the customer (yearly)
Total_calls_made	Total number of calls made by the customer to the bank or customer service (yearly)

Data Overview

The initial steps to get an overview of any dataset are to observe the first few rows to verify data loading, understand the number of rows and columns, check data types, identify missing values and duplicates, and review the statistical summary.

- The dataset contains 660 rows and 7 columns.
- No fully duplicate rows are present, confirming data integrity at the row level.
- The Customer Key column has 5 Customer Keys that appear more than once (10 rows total), but with different attribute values across occurrences. These represent customer profile changes over time and are valid records — they are not removed.
- No missing values are present in any column, so no imputation is required.
- All columns are of int64 datatype. The five meaningful clustering features are Avg_Credit_Limit, Total_Credit_Cards, Total_visits_bank, Total_visits_online, and Total_calls_made.
- SI_No and Customer Key are identifier columns and are dropped before clustering.
- Avg_Credit_Limit has a wide range (1,000 to 200,000), indicating a heavily skewed distribution that will require scaling before clustering.

Exploratory Data Analysis

Each variable is visualized using the most appropriate chart type. Continuous numeric variables use a combined boxplot and histogram with KDE. Discrete or count variables use annotated bar charts with percentage labels. Bivariate analysis uses a correlation heatmap, pairplot, and individual scatter and boxplots to understand inter-feature relationships.

Univariate Analysis

Each variable is individually analysed to understand its distribution, central tendency, spread, and skewness before examining relationships between variables.

Average Credit Limit

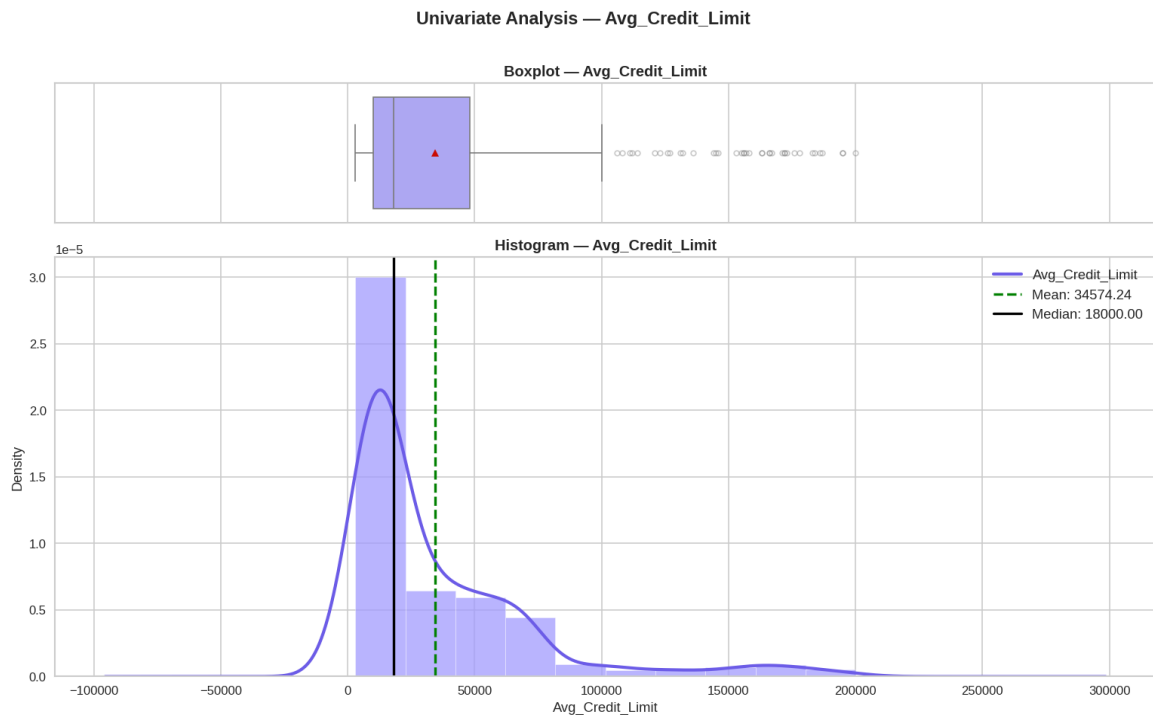


Figure 1: Univariate Analysis — Avg_Credit_Limit (Boxplot + Histogram with KDE)

- The distribution of Avg_Credit_Limit is heavily right-skewed, with the majority of customers having credit limits in the lower range (1,000 – 40,000).
- The mean is significantly higher than the median, confirming the influence of a small number of customers with very high credit limits (up to 200,000).
- The boxplot reveals high-value outliers beyond the upper fence of 105,000. These likely represent premium or High Net Worth Individual (HNI) customers — genuine business segments rather than data errors, and are retained for analysis.
- The wide spread of this variable emphasizes the need for data scaling before clustering to prevent this feature from dominating distance calculations.

Total Credit Cards

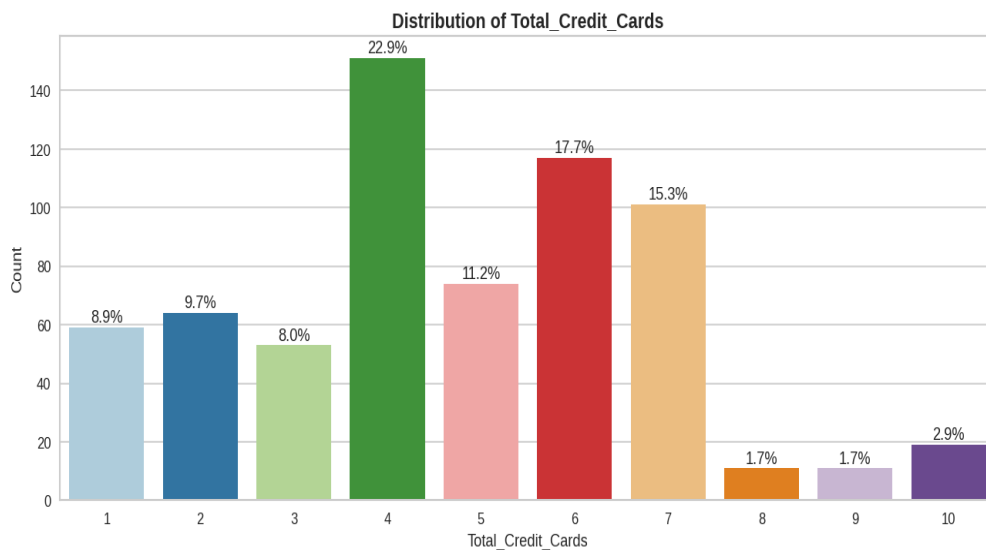


Figure 2: Distribution of Total_Credit_Cards (Annotated Bar Chart)

- The number of credit cards held by customers ranges from 1 to 10, with the most common value being 4 cards (~22.9% of customers).
- The distribution has a slight right skew, with the majority of customers holding between 2 and 7 cards.
- Very few customers hold 8 or more cards (~6%), suggesting that high card counts are relatively uncommon.
- Customers holding a higher number of cards may represent more financially engaged segments with diverse spending patterns.

Total Visits to Bank

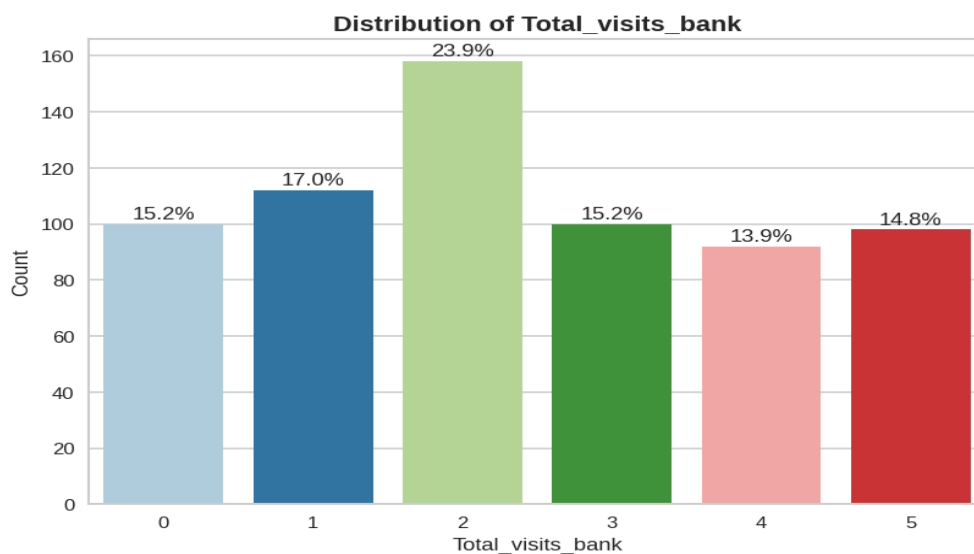


Figure 3: Distribution of Total_visits_bank (Annotated Bar Chart)

- Bank visits range from 0 to 5 per year. A notable 15.2% of customers make no in-person visit at all, reflecting the growing shift toward digital banking.

- The distribution is approximately uniform across all visit counts (0–5), with each category containing roughly 100–158 customers — no single visit count strongly dominates.
- Total_visits_bank has very low skewness (0.142), confirming the near-uniform distribution, unlike the skewed patterns seen in Avg_Credit_Limit and Total_visits_online.
- Customers who visit more frequently (4–5 visits) likely prefer traditional service channels and would benefit most from in-branch service quality improvements.

Total Visits Online

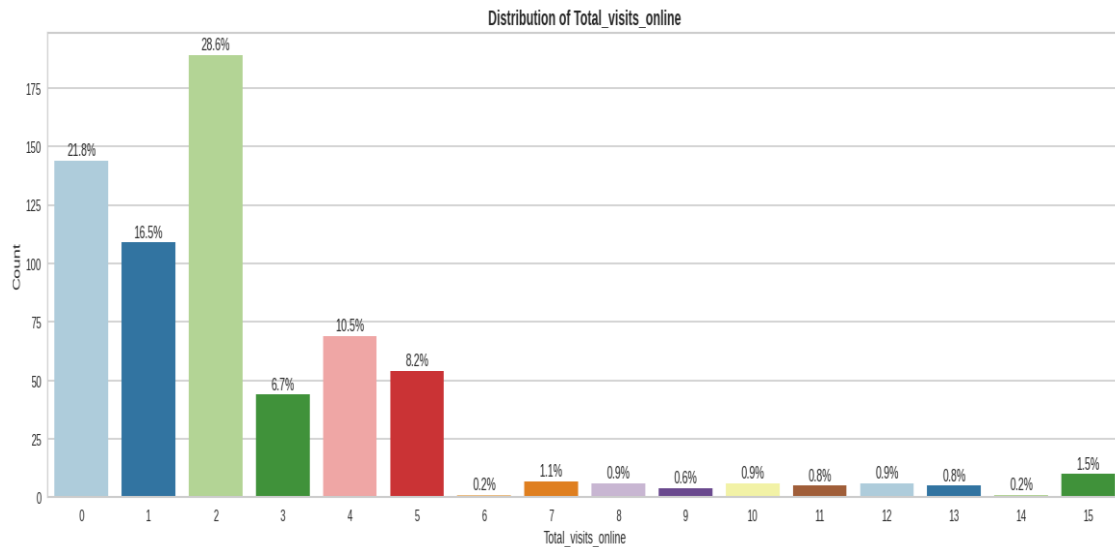


Figure 4: Distribution of Total_visits_online (Annotated Bar Chart)

- Online visits range from 0 to 15 per year. A significant 21.8% of customers make no online visits at all, indicating a substantial digitally-inactive segment.
- The distribution is heavily right-skewed (skewness = 2.226), with the highest frequency at 2 visits (~28.6% of customers).
- The distribution appears bimodal in nature: the majority of customers fall in the 0–5 range, while a smaller but distinct group visits 10–15 times, representing a digitally-intensive segment.
- This variable shows the widest spread among all interaction features (std = 2.94), making it one of the most discriminating variables for segmentation.

Total Calls Made

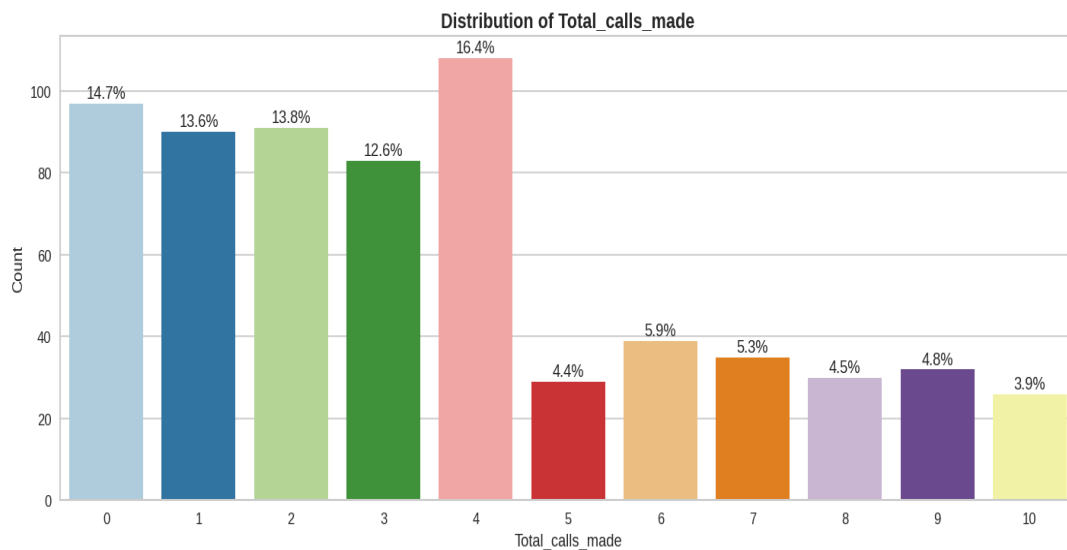


Figure 5: Distribution of Total_calls_made (Annotated Bar Chart)

- Call frequency ranges from 0 to 10 per year, with the highest count at 4 calls (~16.4% of customers).
- The distribution is mildly right-skewed (skewness = 0.659), with the bulk of customers making between 0 and 5 calls per year.
- Approximately 14.7% of customers made no calls at all, suggesting a digitally or branch-engaged group with no service complaints.
- A notable group of customers makes 6 or more calls per year (~24% of all customers), indicating persistent service issues for a significant portion of the base. These customers are the primary motivation for the Operations team's service delivery upgrade.

Bivariate Analysis

The following analyses explore relationships between pairs of variables to understand how they interact and what patterns emerge across the dataset.

Correlation Heatmap

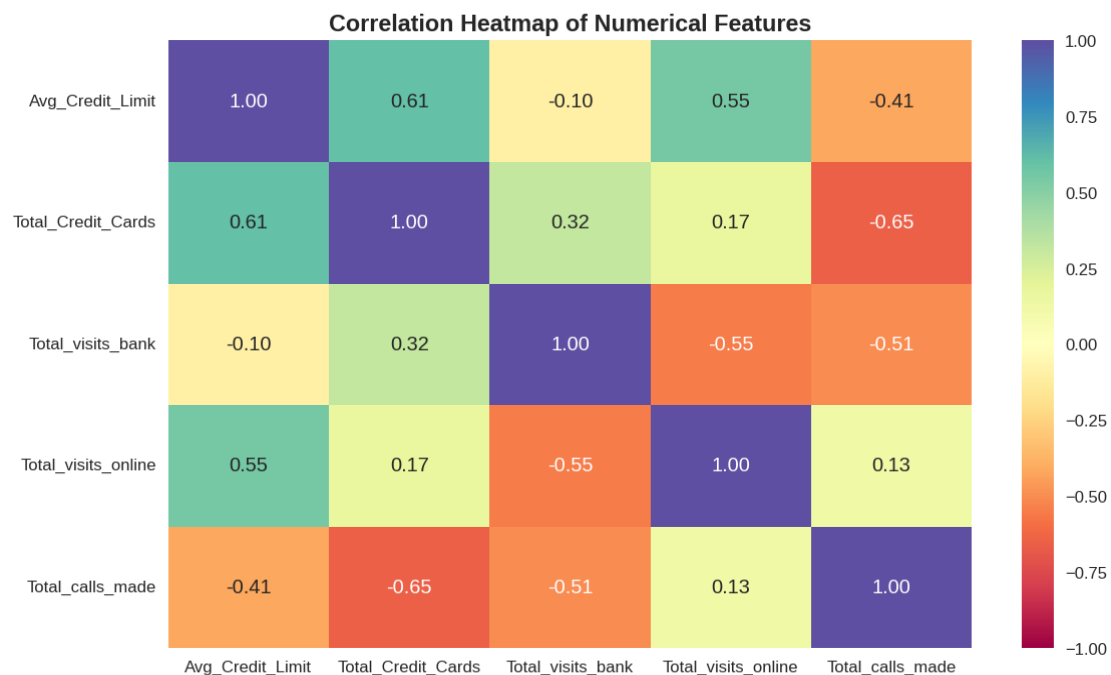


Figure 6: Correlation Heatmap of Numerical Features

- Avg_Credit_Limit and Total_Credit_Cards show a strong positive correlation (0.61), suggesting customers with higher credit limits tend to hold significantly more credit cards — a clear indicator of high financial engagement.
- Total_Credit_Cards and Total_calls_made show a strong negative correlation (-0.65), indicating that customers with more credit cards make far fewer calls — they are likely more financially savvy and self-sufficient.
- Total_visits_bank and Total_visits_online are moderately negatively correlated (-0.55), suggesting that customers who visit the branch frequently tend to use online channels less — indicating two distinct channel preferences.
- Avg_Credit_Limit and Total_calls_made show a moderate negative correlation (-0.41), meaning higher-value customers call less — they may have fewer service pain points or prefer to resolve issues through digital channels.
- Total_visits_bank and Total_calls_made are negatively correlated (-0.51), suggesting that branch visitors resolve their queries in person rather than calling.
- Overall, these correlations reveal strong underlying patterns that will help the clustering algorithm form meaningful, distinct customer groups.

Pairplot of Numerical Features

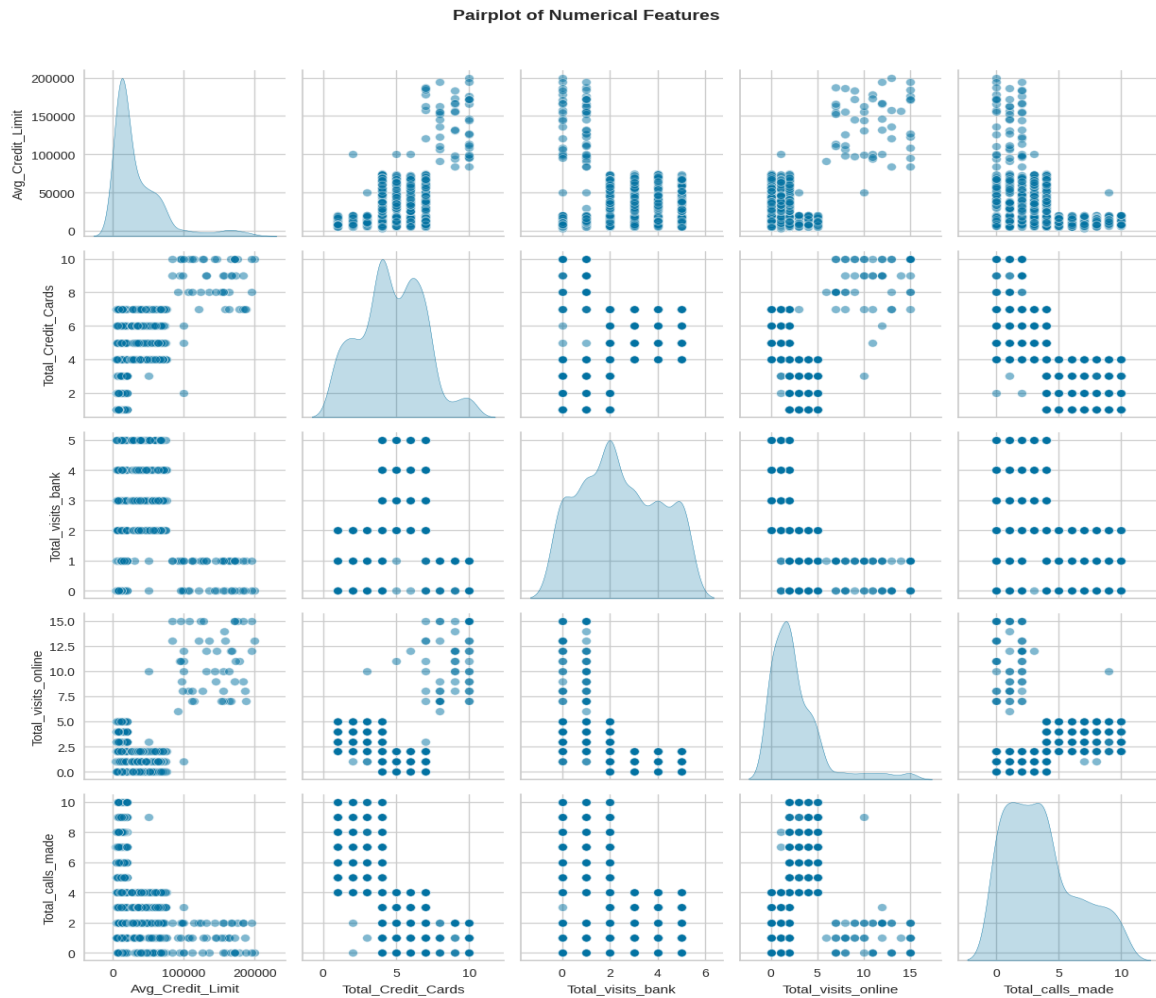


Figure 7: Pairplot of Numerical Features

- The diagonal KDE plots confirm that Avg_Credit_Limit is heavily right-skewed, while the interaction features are more uniformly distributed.
- The scatter plots clearly show a positive relationship between Avg_Credit_Limit and Total_Credit_Cards, and a negative relationship between Total_calls_made and both Avg_Credit_Limit and Total_Credit_Cards.
- The inverse relationship between Total_visits_bank and Total_visits_online is visible — customers tend to prefer one channel over the other.
- There are visible natural groupings of points in several scatter plots, particularly those involving Avg_Credit_Limit, hinting that distinct clusters already exist in the data.

Avg Credit Limit vs Total Credit Cards

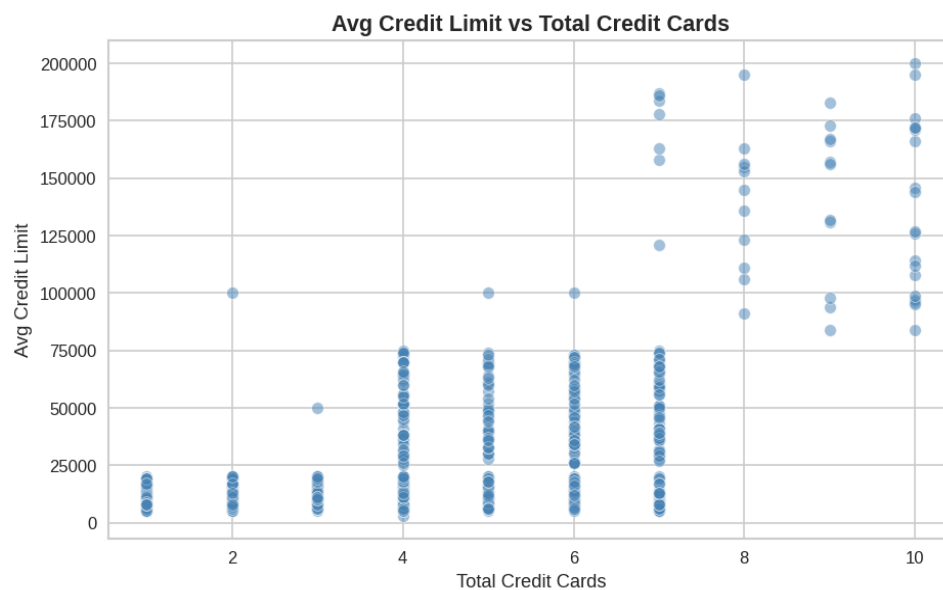


Figure 8: Scatter Plot — Avg Credit Limit vs Total Credit Cards

- There is a clear positive relationship between the number of credit cards held and the average credit limit — customers with more credit cards tend to have significantly higher credit limits.
- Customers with 7 or more credit cards are almost exclusively concentrated in the higher credit limit range (above 50,000), while customers with 1–3 cards are clustered at the lower end.
- This strong positive association (correlation = 0.61) confirms that both variables jointly capture a customer's overall financial engagement and creditworthiness.

Total Credit Cards vs Total Calls Made

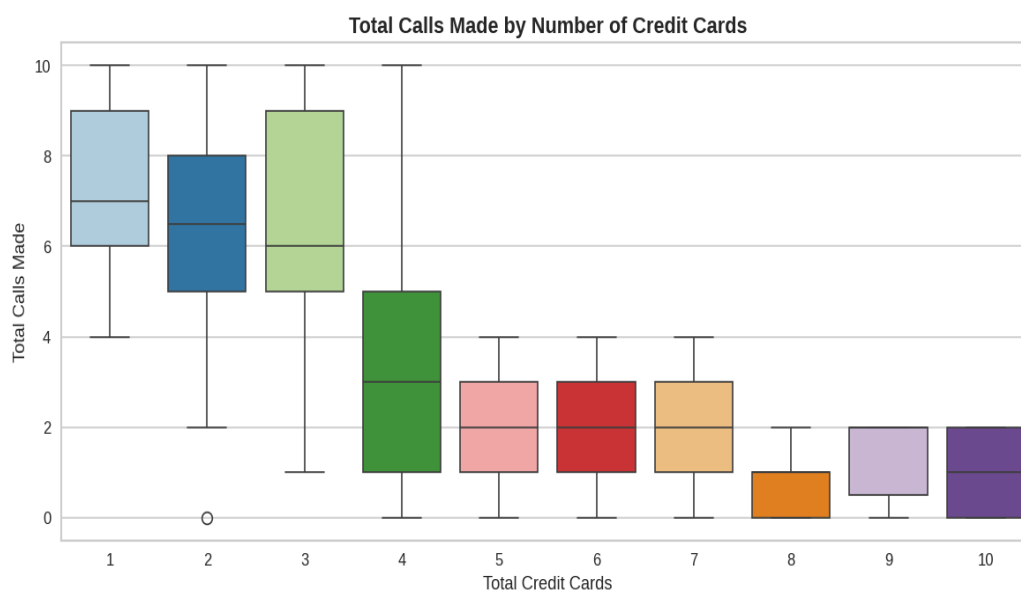


Figure 9: Boxplot — Total Calls Made by Number of Credit Cards

- There is a strong negative relationship (correlation = -0.65) between the number of credit cards and call frequency. Customers holding 1–4 cards have a noticeably higher median call count compared to customers holding 6 or more cards.
- Customers with 5 or more credit cards make significantly fewer calls — their median call count drops to approximately 2 or below, suggesting that higher financial engagement correlates with greater self-sufficiency.
- This pattern implies that low-card customers may be newer, less experienced, or more prone to service issues — making them the primary target for both support improvements and credit card upselling campaigns.

Total Visits Bank vs Total Visits Online

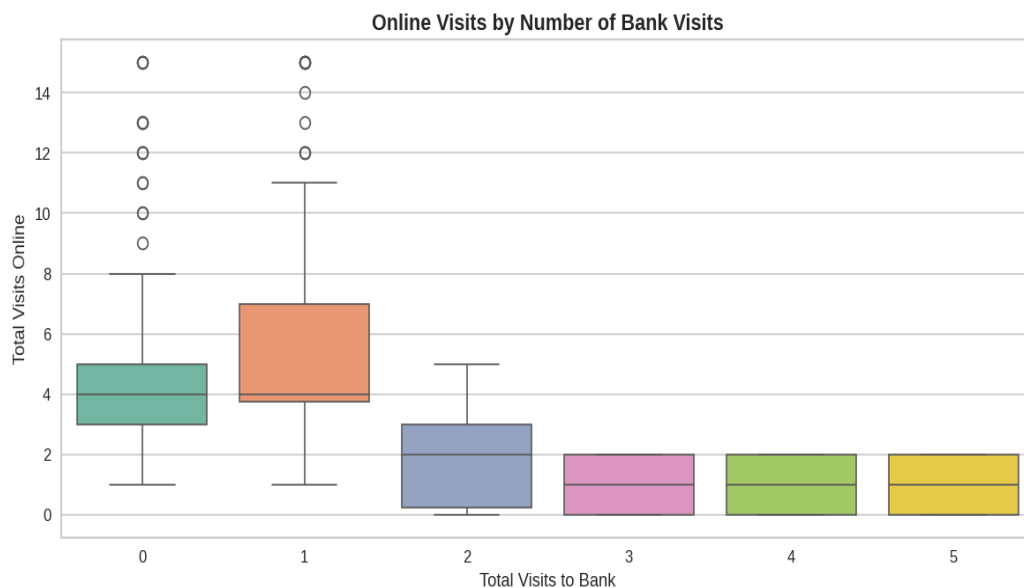


Figure 10: Boxplot — Online Visits by Number of Bank Visits

- There is a clear inverse relationship (correlation = -0.55) between in-branch visits and online visits. Customers who make 0–1 bank visits tend to have significantly higher online visit counts (median ~5), while customers making 3–5 bank visits have very low online activity (median ~1).
- This confirms the existence of two distinct channel preferences in the customer base: a digital-preferring group and a branch-preferring group — a key insight that drives the three-cluster segmentation.
- Very few customers show high activity on both channels simultaneously, reinforcing that these two interaction styles are largely mutually exclusive.

Avg Credit Limit vs Total Calls Made

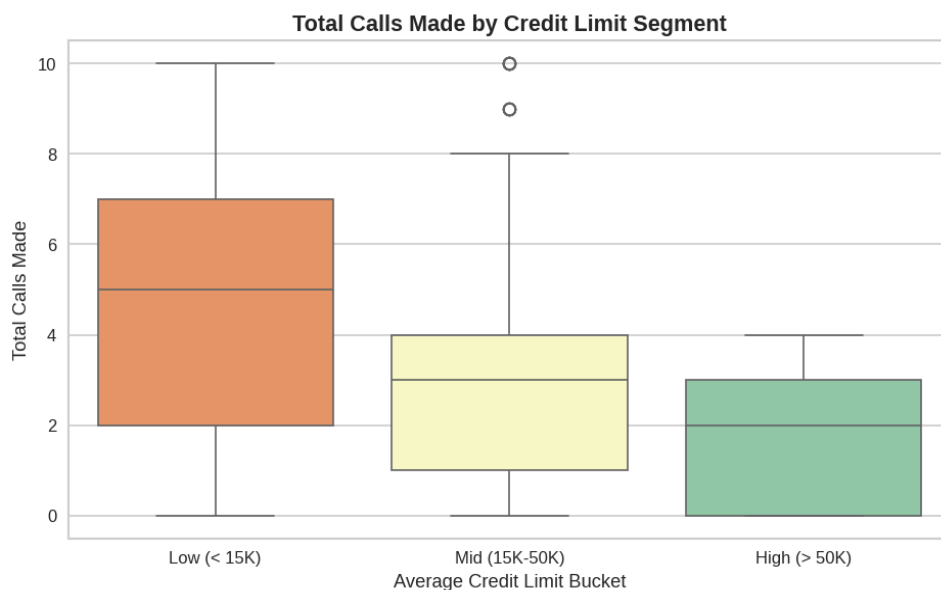


Figure 11: Boxplot — Total Calls Made by Credit Limit Segment

- Customers with low credit limits (< 15,000) have the highest median call frequency (~4–5 calls/year), while customers with high credit limits (> 50,000) make far fewer calls (~1–2 calls/year).
- This strong inverse relationship (correlation = -0.41) suggests that lower-value customers experience more service friction, while higher-value customers are either more self-sufficient or receive better proactive service.
- From a business perspective, this directly identifies the low-credit-limit segment as the primary driver of call center load.

Data Preprocessing

Missing Value Treatment

- The dataset has no missing values in any column. Therefore, no imputation or missing value treatment is required.

Outlier Detection and Treatment

- Avg_Credit_Limit: 39 outliers (~5.9%) detected beyond the upper fence of 105,000. These represent premium / HNI customers — a genuine and valuable segment. Removing them would eliminate an important business group.
- Total_visits_online: 37 outliers (~5.6%) detected above the upper fence of 8 visits. These represent highly digitally engaged customers who interact extensively online. These are meaningful segments, not data errors.
- All other features (Total_Credit_Cards, Total_visits_bank, Total_calls_made): No outliers detected.

Decision: Outliers are retained for this analysis. In a customer segmentation context, extreme values often define distinct and valuable customer segments (e.g., HNI customers, highly digital users). Removing them would deprive the business of potentially the most actionable insights. Additionally, since Standard Scaling is applied before clustering, the effect of outliers on distance-based algorithms is partly mitigated.

Feature Engineering

The dataset already contains well-defined features relevant to the business objective: financial attributes (Avg_Credit_Limit, Total_Credit_Cards) and interaction channel attributes (Total_visits_bank, Total_visits_online, Total_calls_made). No additional feature engineering is required. All five features are directly interpretable and meaningful for customer segmentation without further transformation.

Data Scaling

StandardScaler (Z-score normalization) is applied to all five features before clustering. This transforms each feature to have a mean of 0 and standard deviation of 1. Scaling is essential because Avg_Credit_Limit ranges up to 200,000 while interaction features range from 0 to 15. Without scaling, Avg_Credit_Limit would dominate all distance calculations and render the other features ineffective. Z-score normalization is preferred over Min-Max scaling for clustering as it is less sensitive to extreme outlier values and works well with Euclidean-based distance measures used in both K-Means and Hierarchical Clustering.

K-Means Clustering

K-Means is a centroid-based clustering algorithm that partitions data into K clusters by minimizing within-cluster variance (inertia). It is efficient, scalable, and works well when clusters are roughly spherical. The Elbow Method and Silhouette Score are used to determine the optimal number of clusters.

Elbow Curve

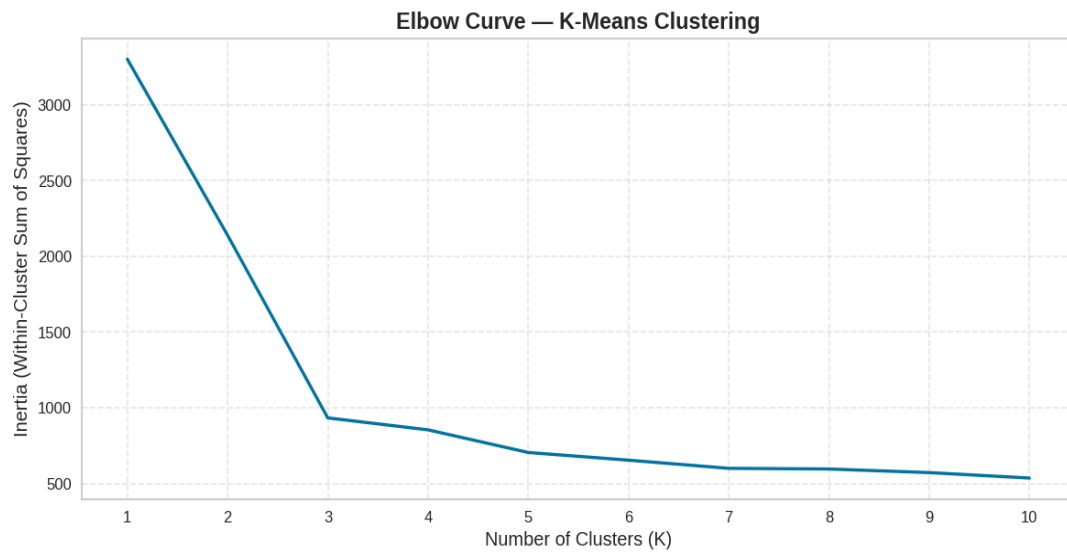


Figure 12: Elbow Curve — K-Means Clustering

- The inertia decreases sharply from K=1 to K=3, and the rate of decrease begins to slow significantly around K=3.
- The curve shows a noticeable elbow or inflection point at K=3, beyond which adding more clusters provides diminishing returns in terms of variance reduction.
- K=2 and K=3 are the candidate values to explore further using silhouette scores.

Silhouette Score Analysis

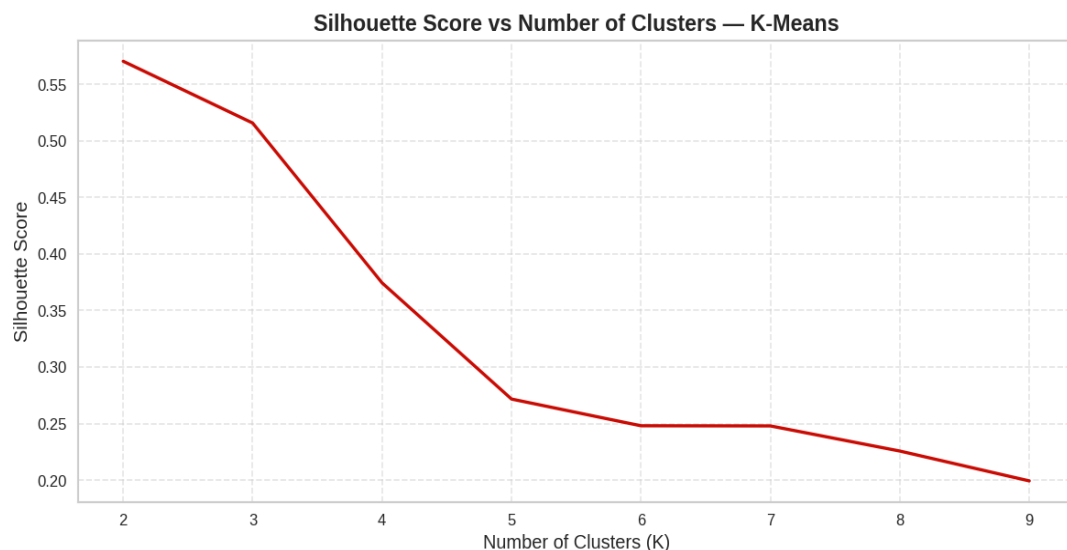


Figure 13: Silhouette Score vs Number of Clusters — K-Means

- The silhouette score measures how similar an observation is to its own cluster compared to other clusters. A higher score indicates better-defined, more separated clusters.
- The highest silhouette score is observed at K=2 (0.5703), followed closely by K=3 (0.5157). Scores decline significantly for K=4 and beyond.

- While K=2 gives the highest silhouette score, two clusters are too broad to capture the meaningful business distinctions required — customers with moderate credit limits and branch-visit preferences would be merged with either the high-value or low-value group.
- K=3 is selected as it offers the best balance between statistical validity and business interpretability, producing three distinct and actionable customer segments.

Silhouette Visualizer

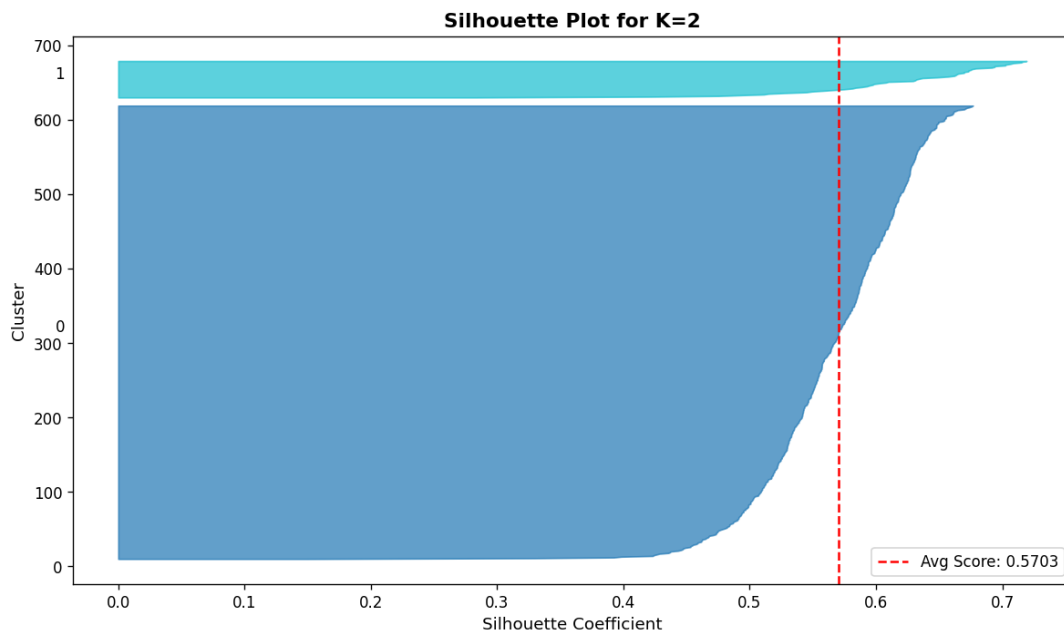


Figure 14: Silhouette Plot for K=2

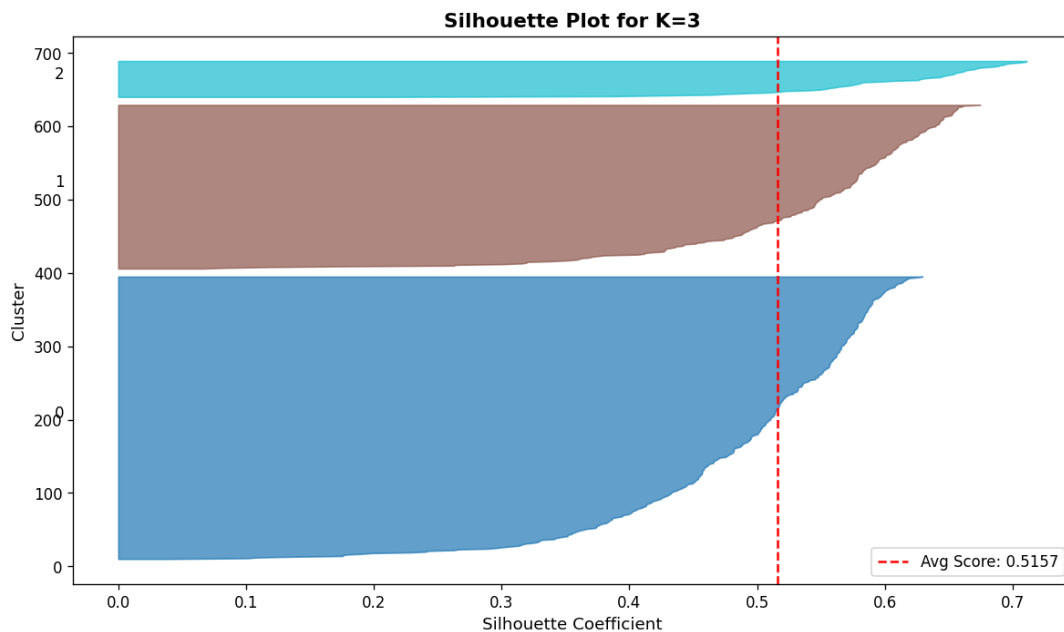


Figure 15: Silhouette Plot for K=3

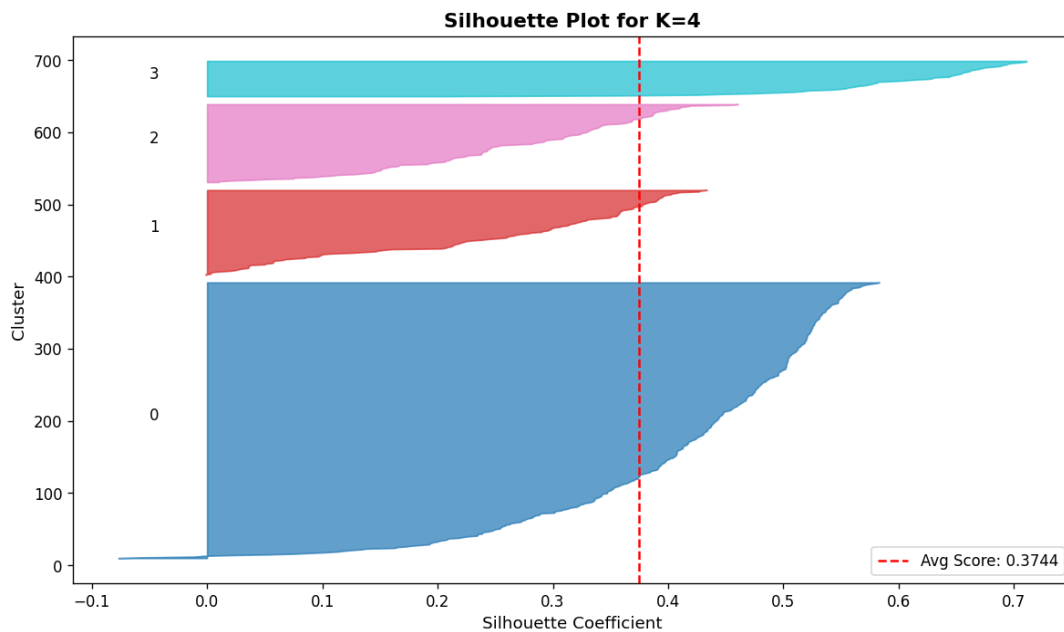


Figure 16: Silhouette Plot for K=4

- At K=2, the average silhouette score is the highest, but the two clusters are very broad and lump together customers with meaningfully different behaviours.
- At K=3, all three clusters show positive silhouette scores with fairly uniform widths. The average score remains high (0.5157) and the three clusters are clearly separated and business-interpretable.
- At K=4, silhouette scores drop noticeably, with some clusters showing very thin profiles — indicating that splitting into 4 groups introduces overlap rather than clarity.

Decision: K=3 is selected as the optimal number of clusters for K-Means, as it provides the best balance between statistical quality and meaningful business segmentation.

K-Means Cluster Profiling

The following visualizations profile the three K-Means clusters across all five features.

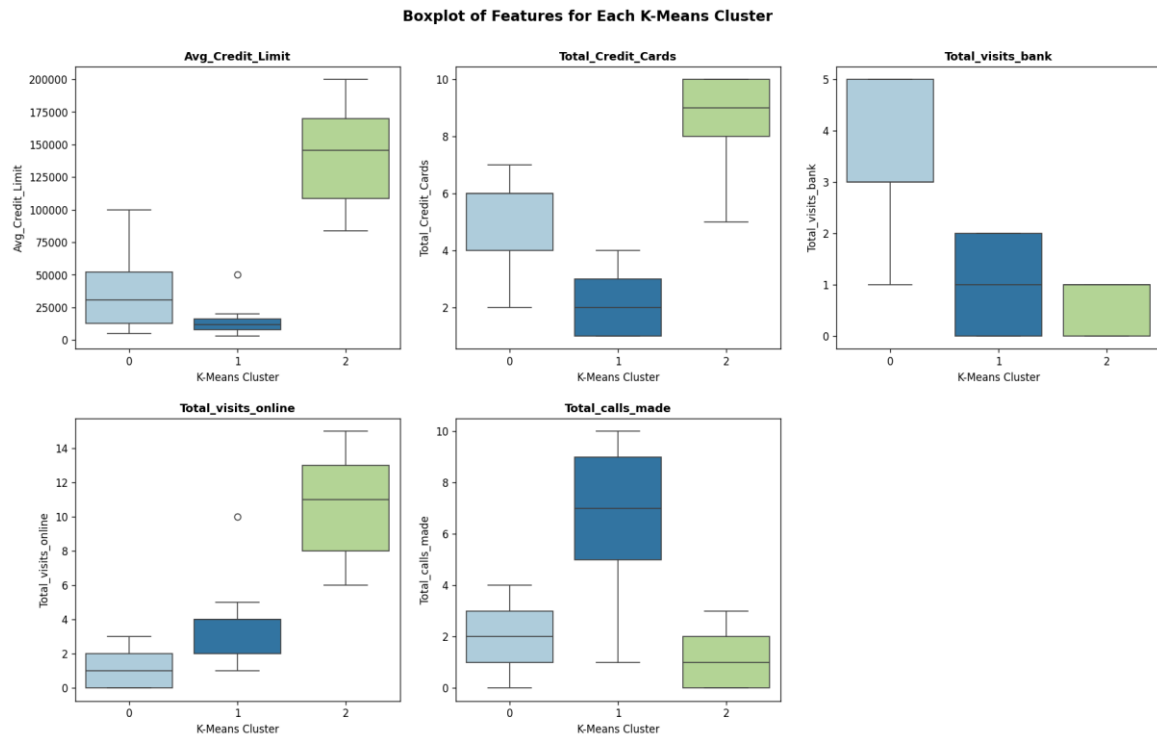


Figure 17: Boxplots of Features for Each K-Means Cluster

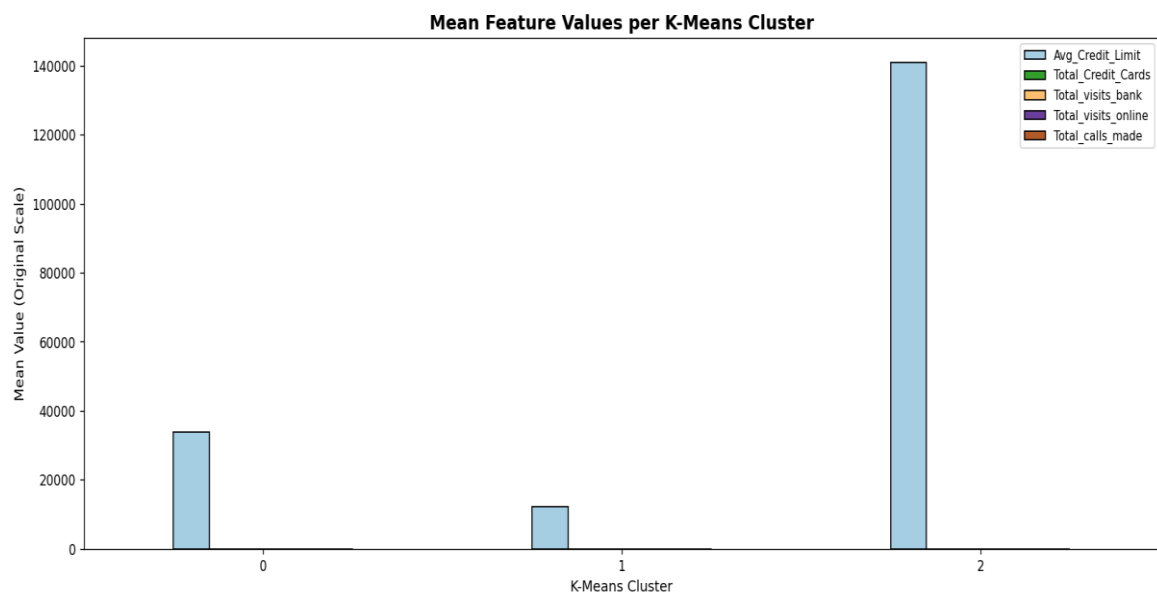


Figure 18: Mean Feature Values per K-Means Cluster

Cluster	Label	Avg Credit Limit	Credit Cards	Bank Visits	Online Visits	Calls Made	Size
0	Mid-Value, Branch-Preferring	~33,800	~5.5	~3.5 (Highest)	~1.0 (Lowest)	~2.0	386 (58%)

1	Low-Value, Call-Heavy	~12,200	~2.4 (Lowest)	~0.9	~3.5	~6.9 (Highest)	224 (34%)
2	High-Value, Digital-First	~141,040 (Highest)	~8.7 (Highest)	~0.6 (Lowest)	~10.9 (Highest)	~1.1 (Lowest)	50 (8%)

- Cluster 0 — Mid-Value, Branch-Preferring (386 customers): Mid-range credit limit (~33,782), highest bank visit frequency (~3.5/year), very low online visits (~1.0/year), and low call frequency (~2.0/year). Every single customer visits the branch at least once per year (0% have zero bank visits), making this segment uniquely reliant on in-person banking.
- Cluster 1 — Low-Value, Call-Heavy (224 customers): Lowest credit limit (~12,174), fewest credit cards (~2.4), and highest call frequency (~6.9/year). Critically, 72.3% of customers in this segment make 6 or more calls per year, and not a single customer makes zero calls — indicating systemic, persistent service dissatisfaction.
- Cluster 2 — High-Value, Digital-First (50 customers): Highest credit limit (~141,040), most credit cards (~8.7), and highest online visit frequency (~10.9/year). 64% make 10 or more online visits per year, 32% make zero calls, and 40% never visit a branch — highly self-sufficient premium customers.

Hierarchical Clustering

Hierarchical Clustering builds a hierarchy of clusters by progressively merging (agglomerative) data points. Unlike K-Means, it does not require specifying K in advance and produces a dendrogram that helps in visually deciding the number of clusters. The Cophenetic Correlation Coefficient is used to evaluate linkage method quality.

Cophenetic Correlation Analysis

The Cophenetic Correlation Coefficient measures how faithfully the dendrogram preserves the pairwise distances between the original data points. A value closer to 1 indicates a better-fit dendrogram.

Linkage Method (Euclidean)	Cophenetic Correlation
Single	0.7391
Complete	0.8600
Average	0.8977 (Highest)
Centroid	0.8939
Ward	0.7415

Weighted	0.8862
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- Average linkage achieves the highest cophenetic correlation (0.8977), meaning its dendrogram most accurately represents the true pairwise distances.
- Ward linkage has a comparatively lower cophenetic correlation (0.7415). However, Ward linkage is well-known for producing more compact and visually distinct clusters with clear separation in the dendrogram, making it more practical for customer segmentation.
- Single linkage has the lowest cophenetic correlation (0.7391) and produces the chaining effect — not suitable for this dataset.

Dendrograms for All Linkage Methods

Dendrograms — All Linkage Methods (Euclidean Distance)

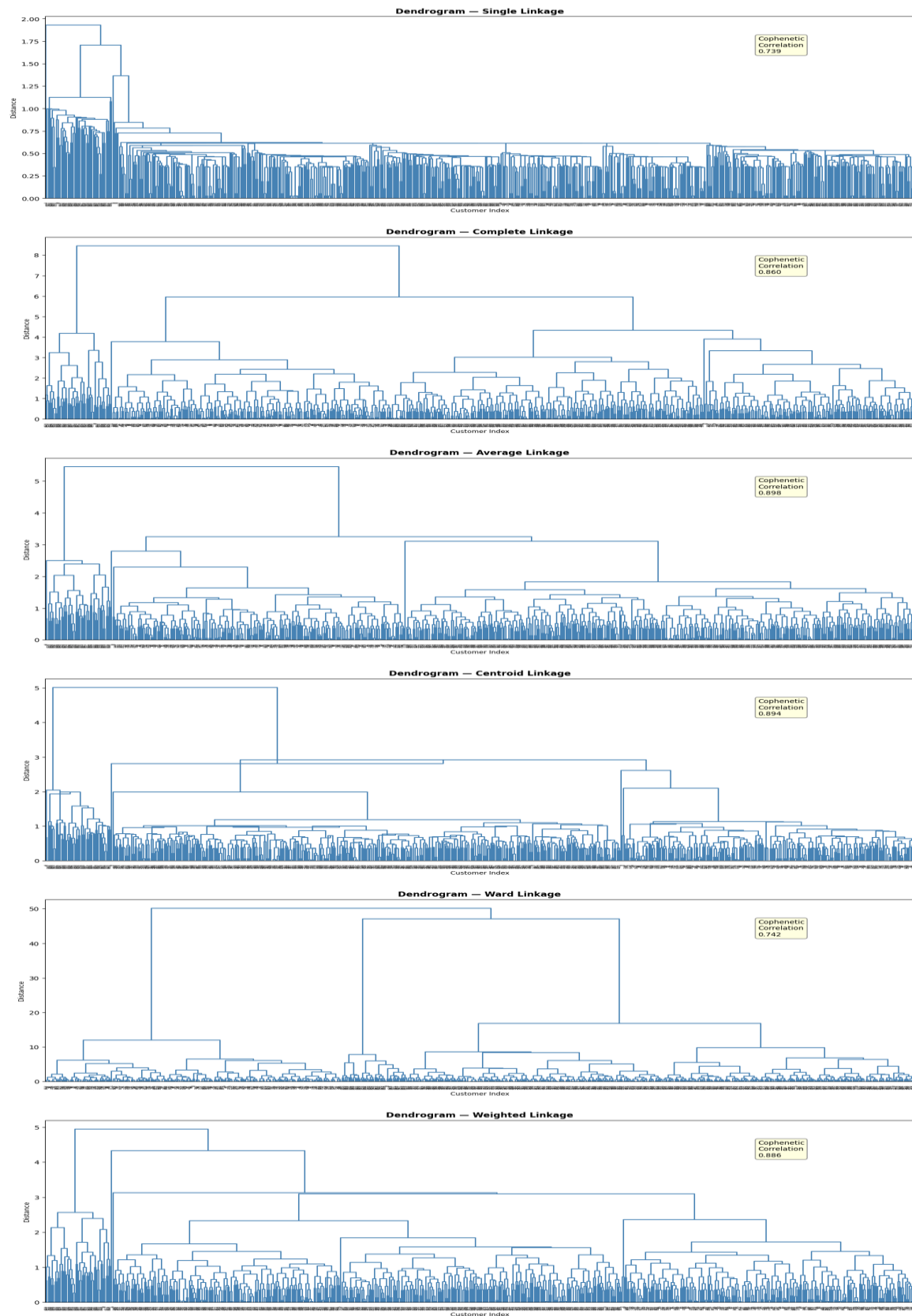


Figure 19: Dendrograms for All Linkage Methods (Euclidean Distance)

- Single Linkage (0.739): Produces a classic chaining effect — most customers are absorbed progressively into one large cluster. Not suitable for this dataset.
- Complete Linkage (0.860): More balanced clusters with reasonable boundaries, but the separation between groups is not very pronounced.
- Average Linkage (0.898): Highest cophenetic correlation; the dendrogram shows reasonably compact clusters. Suggests approximately 3 clusters as a meaningful cut point.
- Centroid Linkage (0.894): High cophenetic correlation; dendrogram structure is similar to average linkage.
- Ward Linkage (0.742): Although cophenetic correlation is lower, it produces the most visually distinct and compact clusters with the clearest gaps between major merges. A horizontal cut clearly suggests 3 clusters.
- Weighted Linkage (0.886): High cophenetic correlation; similar structure to complete linkage.

Decision: Ward linkage is selected for the final hierarchical clustering model, as it produces the most visually distinct and compact clusters.

Ward Dendrogram — Optimal Number of Clusters

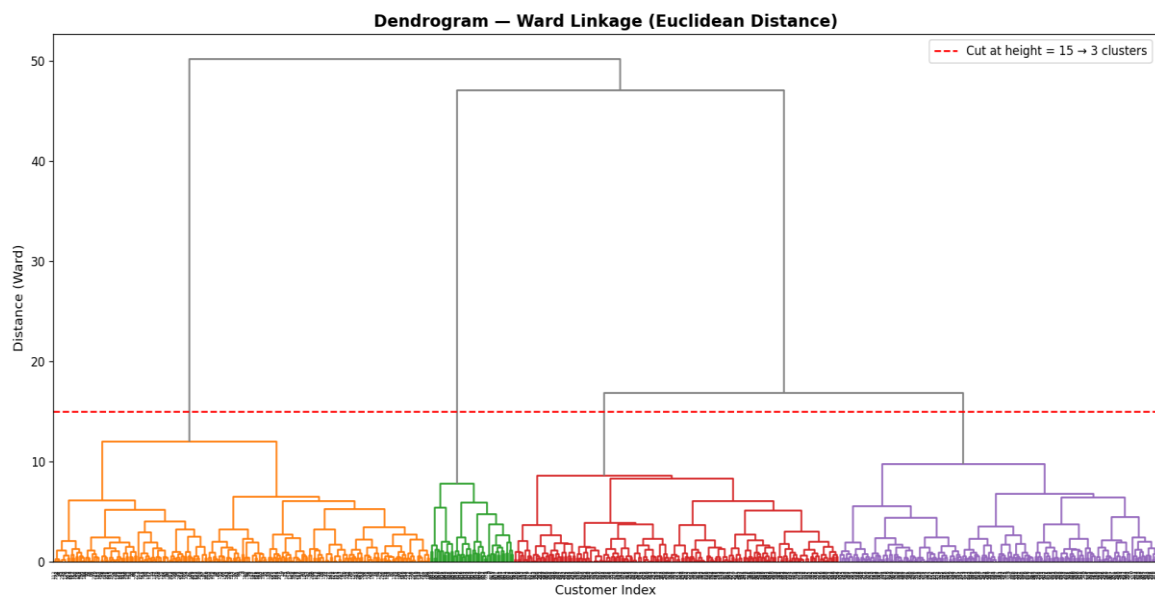


Figure 20: Dendrogram — Ward Linkage with Cut at Height = 15

- The Ward dendrogram shows 3 clearly separated major cluster groups when the tree is cut at an appropriate height. The three groups are visually well-defined with large vertical gaps before the final merge.
- A horizontal cut at approximately height = 15 gives 3 well-separated clusters, consistent with the K-Means findings.
- Beyond 3 clusters, merge distances become significantly smaller, indicating that further splits would produce less meaningful sub-segments.
- K = 3 is therefore selected as the optimal number of clusters for Hierarchical Clustering.

Silhouette Score Validation

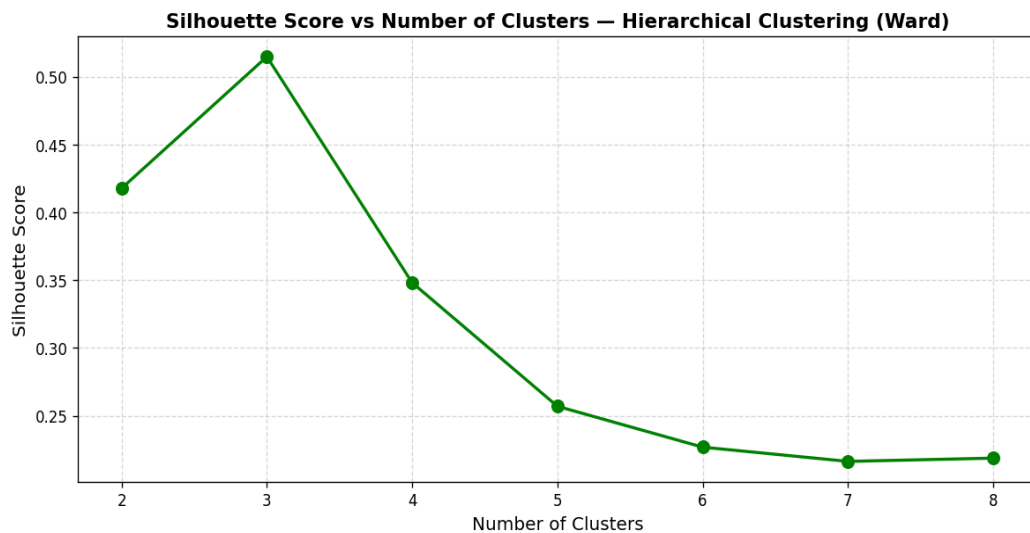


Figure 21: Silhouette Score vs Number of Clusters — Hierarchical Clustering (Ward)

- The silhouette score for Hierarchical Clustering (Ward linkage) peaks at K=3 (0.5148), with K=2 being notably lower (0.4177) — unlike K-Means where K=2 scored highest. This confirms that Ward linkage naturally identifies 3 as the optimal cluster count.
- Scores decline steadily for K=4 and beyond, reinforcing that splitting into more than 3 clusters introduces noise rather than meaningful segments.
- The alignment between the dendrogram visual cut at 3 clusters and the silhouette score peak at K=3 strongly validates 3 as the optimal cluster count for Hierarchical Clustering.

Hierarchical Clustering — Cluster Profiling

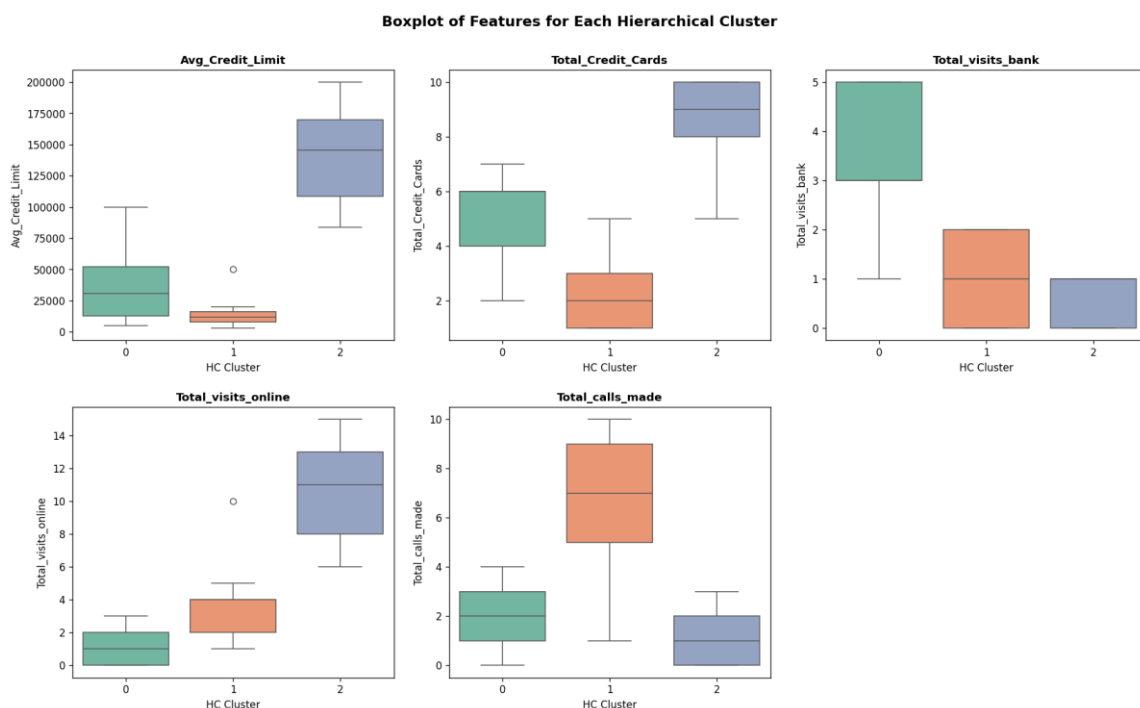


Figure 22: Boxplots of Features for Each Hierarchical Cluster

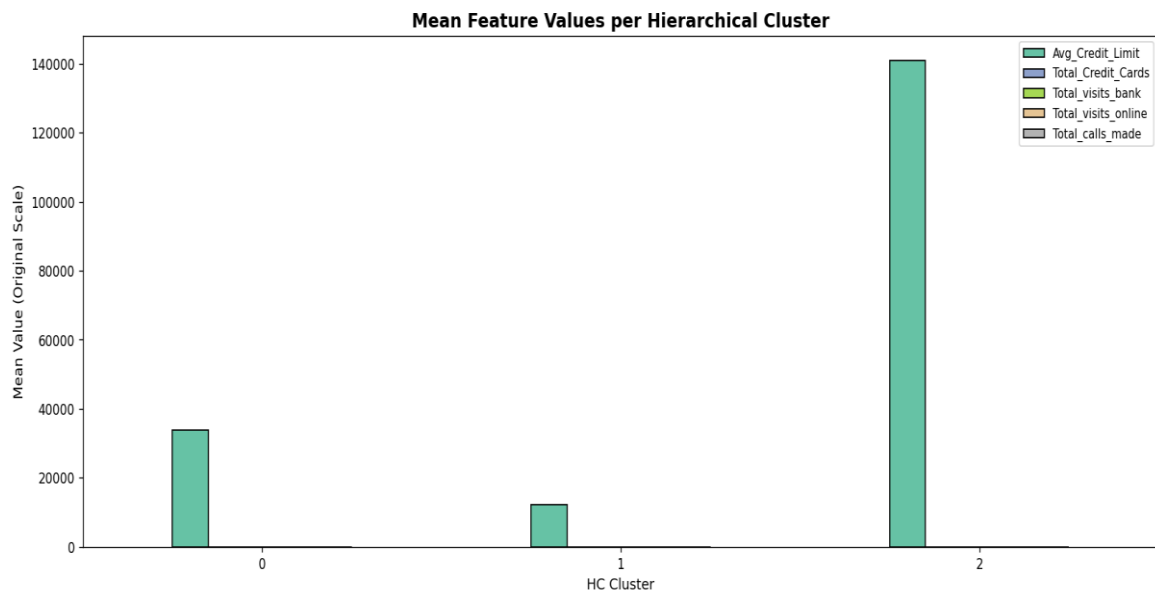


Figure 23: Mean Feature Values per Hierarchical Cluster

The HC cluster profiles are almost identical to the K-Means profiles, further validating the stability of the three segments:

- HC Cluster 0 — Mid-Value, Branch-Preferring (385 customers): Mid-range credit limit (~33,852), highest bank visit frequency (~3.5/year), very low online visits (~1.0/year), and low call frequency (~2.0/year). Corresponds almost exactly to K-Means Cluster 0.
- HC Cluster 1 — Low-Value, Call-Heavy (225 customers): Lowest credit limit (~12,151), fewest credit cards (~2.4), highest call frequency (~6.9/year), moderate online engagement (~3.5/year), and very low branch visits (~0.9/year). Corresponds almost exactly to K-Means Cluster 1.
- HC Cluster 2 — High-Value, Digital-First (50 customers): Highest credit limit (~141,040), most credit cards (~8.7), highest online visit frequency (~10.9/year), fewest bank visits (~0.6/year), and very low call frequency (~1.1/year). Identical to K-Means Cluster 2 — every single customer is assigned to the same cluster by both methods.

K-Means vs. Hierarchical Clustering — Comparison

Having applied both clustering techniques, the results are compared across multiple dimensions to determine the preferred approach for deployment.

Cluster Profile Comparison: K-Means vs. Hierarchical Clustering

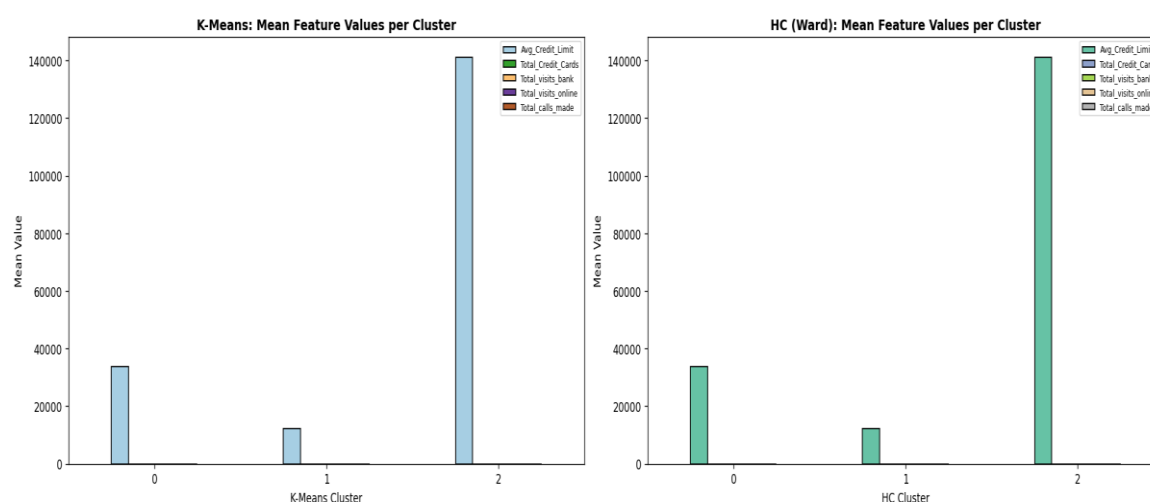


Figure 24: Side-by-Side Cluster Profile Comparison — K-Means vs. Hierarchical Clustering

Criterion	K-Means	Hierarchical Clustering (Ward)
Optimal Clusters	3	3
Silhouette Score	0.5157	0.5148
Agreement (ARI)	—	0.9944 (near perfect)
Cluster Interpretability	High	High (aided by dendrogram)
Scalability	Faster for large data	Slower — $O(n^2)$ memory
Requires K upfront	Yes	No — dendrogram guides selection
Sensitivity to initialization	Yes (mitigated by random_state)	No

- Both algorithms converge on 3 as the optimal number of clusters, providing strong mutual validation. The Adjusted Rand Index of 0.9944 confirms near-perfect agreement between the two solutions — only 1 customer is assigned differently between the two methods.
- The three segment profiles — Low-Value/Call-Heavy (Cluster 1), Mid-Value/Branch-Preferring (Cluster 0), and High-Value/Digital-First (Cluster 2) — emerge identically from both methods, confirming these are genuine, stable customer segments in the data.
- K-Means yields a marginally higher silhouette score (0.5157 vs 0.5148), but the difference is negligible. Both methods are essentially equivalent in cluster quality.
- Hierarchical Clustering provides the added advantage of the dendrogram, offering a visual, assumption-free way to determine cluster count. Notably, the HC silhouette score peaks at K=3 (not K=2 like K-Means), making the cluster count decision more straightforward with HC.

- K-Means is preferred for deployment given its faster execution time and ease of assigning new customers to clusters using trained centroids. However, both methods yield fully equivalent and actionable business insights.

Actionable Insights and Recommendations

Both clustering techniques consistently and reliably identify 3 meaningful customer segments. The following insights and recommendations address the dual business objectives of improving marketing effectiveness and upgrading service delivery.

Segment Summary

Segment 1 — Low-Value, Call-Heavy Customers (Cluster 1 — 34% of customers)

Profile: These customers have the lowest credit limits (all under 50,000, with 96% under 25,000), fewest credit cards (1–4 cards only), and make by far the most calls to the bank (~6.9/year). Critically, 72.3% of customers in this segment make 6 or more calls per year, and not a single customer makes zero calls — indicating systemic, persistent service dissatisfaction rather than isolated incidents. They use online channels moderately (all make at least 1 online visit) but 35.7% never visit a branch.

Marketing Recommendations

- Run targeted campaigns for credit limit enhancement and entry-level credit card upgrade offers to grow this segment's financial relationship with the bank. Moving customers from 1–2 cards to 3–4 cards could be a meaningful first step.
- Offer educational content (FAQs, how-to videos, chatbot assistance) to help them become more self-sufficient and reduce call dependency.
- Use loyalty incentives (cashback, reward points) to deepen engagement and improve product holding.
- Since all customers in this segment use online channels (minimum 1 visit, median 4), digital nudges, in-app prompts, and SMS campaigns are viable and cost-effective reach mechanisms.
- Consider gamified financial literacy programs to help this segment better understand and use their credit products — reducing confusion-driven calls while building brand loyalty.

Operations Recommendations

- Prioritize this cluster for the service delivery model upgrade — 72.3% make 6+ calls/year, making them the dominant driver of call center load.
- The fact that zero customers in this segment make zero calls is a critical signal — the service gap is universal within this group. A root cause analysis of the most frequent call types for this segment should be conducted immediately.
- Implement proactive outreach (SMS/email alerts for billing cycles, payment reminders, limit utilization alerts) to resolve common triggers before customers need to call.

- Introduce IVR/chatbot self-service options tailored to this segment's most frequent query types — billing, payment, and limit-related queries are likely dominant.
- Assign dedicated customer care representatives with faster escalation paths and first-call resolution targets.
- Track First Call Resolution (FCR) rate specifically for this segment as a key operations KPI.

Segment 2 — Mid-Value, Branch-Preferring Customers (Cluster 0 — 58% of customers)

Profile: The largest segment at 58% of the customer base. Every single customer visits the branch at least once per year (0% have zero bank visits) — a uniquely defining characteristic. Mid-range credit limits (5,000 to 100,000), 4–7 credit cards, and 37.3% have never used online banking. Their call frequency is low and capped at 4 calls maximum — not a service-distressed group. With an average of 6.47 total interactions per year, these customers engage primarily through the branch.

Marketing Recommendations

- Roll out in-branch personalized offers — these customers are most receptive to face-to-face engagement, and the guaranteed annual branch visit provides a reliable touchpoint for every customer in this segment.
- Cross-sell products such as fixed deposits, insurance, or investment plans during branch visits, as these interactions provide a captive and receptive audience.
- Introduce relationship manager assignments to deepen personal connections and grow product holding — this segment's moderate credit limit range (5K–100K) suggests significant room for credit limit growth.
- Since 37.3% have never used online banking, there is a substantial untapped digital adoption opportunity. Incentivize first digital transactions through branch staff recommendations.
- Design branch-triggered digital onboarding campaigns: when a customer visits the branch, staff can register them for online banking on the spot, converting passive branch visitors into omnichannel customers.

Operations Recommendations

- Since 100% of this segment visits the branch at least once per year, branch experience quality directly determines satisfaction for this entire group. Any deterioration in branch service will impact 58% of the customer base simultaneously.
- Ensure branch staffing levels and service quality are consistently optimized — this is the highest-priority operational lever for this segment.
- Invest in branch ambience and queue management systems to reduce wait times and improve the in-person experience.
- Train branch staff to proactively cross-sell and identify upsell opportunities during every visit, given the guaranteed annual touchpoint.
- Monitor branch Net Promoter Score (NPS) specifically for this segment as the primary satisfaction metric.

Segment 3 — High-Value, Digital-First Customers (Cluster 2 — 8% of customers)

Profile: The smallest but most valuable segment. Every customer has a credit limit of at least 84,000, with 78% exceeding 100,000 and an average of 1.41 lakh. 96% hold 7 or more credit cards, and 64% make 10 or more online visits per year. Remarkably, 32% make zero calls to the bank and 40% never visit a branch — almost entirely self-sufficient. This segment, though only 8% of customers, almost certainly contributes a disproportionately large share of the bank's revenue.

Marketing Recommendations

- Focus on upselling premium credit card products (co-branded cards, travel rewards, concierge services, airport lounge access, forex benefits) — with 96% already holding 7+ cards, the pitch should be on upgrading card tiers, not adding more cards.
- Offer exclusive digital banking features such as higher credit lines, wealth management products, portfolio tracking tools, and priority banking app features.
- Design highly personalized digital campaigns using spending pattern data (in-app notifications, email personalization) — this is the only segment where digital-only marketing is sufficient.
- Introduce a premium loyalty program with tiered rewards and exclusive privileges (travel upgrades, dining benefits, priority customer service) to deepen retention.
- Explore wealth management and investment product cross-sells — customers at this credit limit level are likely to have investable surplus and may respond to curated financial planning offers.

Operations Recommendations

- Ensure the digital banking platform is robust, fast, secure, and feature-rich — for 40% of this segment, the app is their sole interaction channel with the bank. Any downtime or poor UX directly risks churn.
- Offer a dedicated priority helpline (near-zero wait times, dedicated relationship managers) as a fallback — when a high-value customer does call, the experience must be seamless.
- Provide proactive digital notifications (spending insights, credit limit review prompts, investment opportunities, reward point summaries) to add value beyond transactional banking.
- Implement churn early-warning indicators for this segment — even losing 5–10 of these 50 customers could have a meaningful revenue impact. Monitor for declining online visit frequency or increasing call volume as early distress signals.

Overall Strategic Recommendations

1. **Segment-Specific Communication Strategy:** Adopt a strict channel-specific marketing approach — digital-only for Cluster 2, in-branch for Cluster 0, and blended digital plus assisted service for Cluster 1. Sending branch-focused campaigns to Cluster 2 or digital-only campaigns to Cluster 0 will be ineffective and wasteful.
2. **Reduce Call Center Load Immediately:** Cluster 1 (~34% of customers) generates nearly all inbound call volume — 72.3% of this segment makes 6+ calls/year and no one makes zero

calls. Proactive communication, self-service tools, and first-call resolution improvements will directly address the Operations team's service delivery mandate and reduce operational costs significantly.

3. **Protect and Grow High-Value Customers:** Cluster 2 (~8% of customers) likely represents a disproportionate share of the bank's revenue. The digital platform is their primary or sole relationship channel for 40% of them — any degradation in digital service quality poses an immediate churn risk. Retention investments here yield the highest ROI.
4. **Leverage the Guaranteed Branch Touchpoint for Cluster 0:** Every single customer in Cluster 0 visits the branch at least once per year, making it the only segment with a guaranteed annual face-to-face interaction. This is a uniquely valuable cross-sell and upsell window that should be systematically leveraged through trained branch staff and structured conversation guides.
5. **Bridge the Digital Gap for Cluster 0:** 37.3% of this segment has never used online banking. A phased digital adoption strategy — incentivized through in-branch onboarding — will reduce long-term branch operating costs and move customers toward omnichannel engagement.
6. **Revenue Prioritization Framework:** While Cluster 2 is smallest by count, it should receive the highest per-customer investment. Cluster 0 should receive the highest total investment given its size (58%). Cluster 1 requires urgent operational investment to reduce costs and improve satisfaction before marketing investment can be effective.